

01 rough while reading Learn Reinforcement Learning

Saturday, September 19, 2026 1:36 PM

Policy

$$\pi(a|s)$$

$$= p(A_t = a | S_t = s)$$

Example

$$\pi(\text{Right} | 1) = 0.8$$

$$p(A_t = \text{Right} | S_t = 1) = 0.8$$

$$\pi(\text{Left} | 1) = 0.2$$

4.
 $A = \{a_1, a_2, a_3\}$
set of actions

$$\pi(a_1 | s) = 0.1$$

$$\pi(a_2 | s) = 0.7$$

$$\pi(a_3 | s) = 0.2$$

$$\sum_{a \in A} \pi(a|s) = 1$$

$$0 < \pi(a|s) < 1$$

$$\pi(a|s) = p(A=a|S=s)$$

6. Example

Robot
 s = obstacle
at front

$$A = \{\text{Left}, \text{Right}, \text{Forward}\}$$

$$\pi(\text{Left} | s) = 0.6$$

$$\pi(\text{Right} | s) = 0.4$$

$$\pi(\text{Forward} | s) = 0$$

$$A_t \sim \pi(\cdot | s)$$

$$A_t \sim \underbrace{\pi(\cdot | S_t)}_{\text{sample the action}}$$

$$A_t \sim \pi(\cdot | S_t)$$

Sample the action from policy distribution

7.

Stochastic policy
 $A_t \sim \pi(\cdot | S_t)$

8.

9.

Exploitation:
 what believed is best
 $a = \arg \max Q(s, a)$

Exploration:
 Try alternative

$$\pi(a_1 | s) = 0.8$$

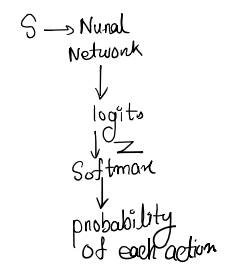
$$\pi(a_2 | s) = 0.2$$

agent mostly exploits a_1
 but sometimes explores a_2

11

$$s = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \end{bmatrix}$$

state as a vector



if 3 actions
 network produces

if 3 action
network produce
 $z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$

Softmax:

$$\pi_{\theta}(a_i | s) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

for given state s
prob of action = a

$\theta \rightarrow$ parameters of network

$\pi_{\theta}(a | s)$
depends on θ
train θ to get good decisions

Coding

15

$$\pi_{\theta}(a | s)$$

policy optimization
by training θ

17

$$s_t = \underbrace{[h_1(t), h_2(t)]}_{\text{channel condition}}, \underbrace{[I_1, I_2]}_{\text{Interference}}$$

$$a_t = \text{action} = p_t$$

$$p_t \in [0, 1]^W$$

$$A = \begin{Bmatrix} 0 & 0.2 & 0.4 & 0.6 \\ & 0.8 & 1 & \end{Bmatrix}$$

$$\pi(a | s)$$

18

18

$\pi(a|s) \rightarrow$ doesn't
tell which
action is
good

$V^\pi(s)$ $Q^\pi(s,a)$
Value functions

21 RL family
Value based
 $\pi(s) = \arg \max Q(s,a)$

Policy based
 $\pi_\theta(a|s)$
Reinforce

Action critic
 $\pi_\theta(a|s)$
 $V_\phi(s)$ on $Q(s,a)$ } learn

Summary:
 $\pi(a|s)$

$$\sum \pi(a|s) = 1$$

$A_t \sim \pi(\cdot|s_t)$
Stochastic

RL lesson 3
Value function
 $V^\pi(s)$

$$S_t \rightarrow A_t \rightarrow R_{t+1}, S_{t+1}$$

$$\pi(a|s)$$

agent wants long
term reward

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$G_t = R_{t+1} + \gamma G_{t+1}$$

Now I am in s ,
how good is
being there

$$V^\pi(s) = \mathbb{E}_\pi [G_t | S_t = s]$$

in state s ,
we have many
possible future.

What's the
expected G given
 s

$$V^\pi(s) = \mathbb{E}_\pi [G_t | S_t = s] \\ = \sum \pi(a|s) G_t$$

3.

two policy
 $\pi_1 \rightarrow$ toward goal
 $\pi_2 \rightarrow$ random move

$$V^{\pi_1}(s) \neq V^{\pi_2}(s)$$

Value function
depends on state
and policy

4

0 1 2 3 4
Goal

moving right, $R = -1$
goal $R = +10$
 $\gamma = 0.9$

if $\pi(s) = \text{Right}$ everywhere
we are at state 0
future $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

if $\pi(s) = \text{Right}$ everywhere
 we are at state 0
 future $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

Reward -1 -1 -1 10

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$= (-1) + 0.8(-1) + 0.8^2(-1) + 0.8^3 10$$

$$G_t = 4.58$$

$$V^\pi(0) = E_{\pi} [G_t | \text{state}=0]$$

only one
 G_t no
 other
 possibility

$$= G_t = 4.58$$

5. Now
 at state 0,
 choose Right

but environment
 20% chance
 of slipping

$$P(S_{t+1}=1 | S_t=0, A_t=R) = 0.8$$

$$P(S_{t+1}=0 | S_t=0, A_t=R) = 0.2$$

many future

0 1 2 3 4

0 0 1 2 3 4

0 0 0 1 2 3 4

if
 $G_t = 4.58$, getting the
 prob = 0.8

$G_t = 3.12$, " " " " = 0.2

$$V^\pi(0) = 0.8(4.58) + 0.2(3.12) = 4.288$$

$$V^\pi(0) = E[G_t | S_t=0]$$

(*)

$$V^\pi(s) = E[G_t | S_t=s]$$

(*)

$$V^{\pi}(s) = E[G_t | S_t = s]$$

$$= \sum_g P(G_t = g | S_t = s)$$

there may many possible future, many possible g with there probability

~~#~~ 7

state = s

r_1, r_2, r_3 —
many possible trajectory

each has
 $G(r_1)$ $G(r_2)$ $G(r_3)$

8

$$V^{\pi}(s) = E_{\pi}[G_t | S_t = s]$$

$$G_t = R_{t+1} + \gamma G_{t+1}$$

$$V^{\pi}(s) = E_{\pi}[R_{t+1} + \gamma G_{t+1} | S_t = s]$$

$$= E_{\pi}[R_{t+1} | S_t = s] + \gamma E_{\pi}[G_{t+1} | S_t = s]$$

Now,

$$V^{\pi}(s') = E_{\pi}[G_{t+1} | S_{t+1} = s']$$

Now,

$$E_{\pi}(G_{t+1} | S_t = s)$$

$$= E_{\pi}\left[E_{\pi}[G_{t+1} | S_{t+1}] \mid S_t = s\right]$$

(law of total expectation)

So,

$$V^{\pi}(s) = E_{\pi}[R_{t+1} | S_t = s]$$

$$+ \gamma E_{\pi}[G_{t+1} | S_t = s]$$

$$= E_{\pi}[R_{t+1} | S_t = s] + \gamma E_{\pi}\left[E_{\pi}[G_{t+1} | S_{t+1}] \mid S_t = s\right]$$

$$= E_{\pi}[R_{t+1} | S_t = s] +$$

$$+ \gamma E_{\pi} [E_{\pi}[R_{t+1}|S_{t+1}] | S_t = s]$$

$$= E_{\pi}[R_{t+1}|S_t = s] + \gamma E_{\pi}[V^{\pi}(S_{t+1})|S_t = s]$$

$$V^{\pi}(s) = E_{\pi}[R_{t+1} + \gamma V^{\pi}(S_{t+1}) | S_t = s]$$

This is Bellman expectation equation

??

law of total expectation

$$E(X) = E[E(X|Y)]$$

$$\begin{aligned} \text{proof:} \\ E[X] &= \sum_{\alpha} \alpha P(X=\alpha) \\ &= \sum_{\alpha} \alpha \sum_y P(X=\alpha, Y=y) \end{aligned}$$

$$= \sum_{\alpha} \alpha \sum_y P(X=\alpha|Y=y) P(Y=y)$$

$$= \sum_y \left(\underbrace{\sum_{\alpha} \alpha P(X=\alpha|Y=y)}_{E(X|Y=y)} \right) P(Y=y)$$

$$= \sum_y E(X|Y=y) P(Y=y)$$

$$= E[E(X|Y)]$$

Bellman
Expectation
Equation

$$V^{\pi}(s) = E_{\pi} \left[\begin{array}{c} R_{t+1} + \gamma V^{\pi}(S_{t+1}) \\ | S_t = s \end{array} \right]$$

9
Current value = immediate
reward
+ γ future value
discounted

10
State 0 $\rightarrow$ Right $\rightarrow$ State 1
prob = 0.8
|
 $\rightarrow$ Stay $\rightarrow$ State 0 prob = 0.2
let immediate reward = -1

$$V^{\pi}(0) = E[R_{t+1}|S_t = 0]$$

$$+ \gamma E[V^{\pi}(S_{t+1})|S_t = 0]$$

$$\begin{aligned}
 V'(0) &= E(R_{t+1} | S_t=0) \\
 &+ \gamma E(V_{\pi}(S_{t+1}) | S_t=0) \\
 &= (-1) + 0.9 \left[0.8 V^{\pi}(1) + 0.2 V^{\pi}(0) \right]
 \end{aligned}$$

Python code

$$G_t = R_{t+1} + \gamma G_{t+1}$$

in state = 0
 my rewards = [-1 -1 -1 10]
 Start iteration from here

$G_4 = 0$ $R_4 = 10$
 $G_3 = \checkmark$
 $G_2 = \checkmark$
 $G_1 = \checkmark$
 $G_0 = \checkmark$

12

$$\hat{V}^{\pi}(0) = \frac{1}{N} \sum G_0^{(i)}$$

N large,
 $\hat{V}^{\pi}(0) \rightarrow V^{\pi}(0)$

the basic idea of
 Monte Carlo estimation
 of a value function

13 Value is not
 Reward

$$R_{t+1} = 10 \leftarrow \text{immediate reward}$$

$$\begin{aligned}
 V^{\pi}(s) &= 50 \rightarrow \\
 &= E(G_t | S_t = s)
 \end{aligned}$$

expected total
 future return
 from state = s

$$V^{\pi}(s)$$

start at s
 follow π
 environment evolve
 collect reward
 find G_t
 average over all
 possible outcome

$V^{\pi}(s)$ = expected
 return when
 following π

16
 $\gamma =$ discount channel
 condition

$$V^\pi(s)$$

17
 π^{bad}

$$V^{\pi^{\text{bad}}}(s)$$

$$\pi^{\text{good}} \quad V^{\pi^{\text{good}}}(s)$$

18
 $\pi(a|s)$

$$R_{t+1} \rightarrow \text{Reward}$$

$$G_t \rightarrow \text{Return}$$

$$V^\pi(s) \rightarrow \text{Value}$$

$$V^\pi(s) = E_\pi(G_t | S_t = s)$$

Bell:

$$V^*(s) = E_\pi(R_{t+1} + \gamma V^*(S_{t+1}) | S_t = s)$$

How do we get
 future value

$$0 \quad 1 \quad 2 \quad 3 \quad 4 \quad \text{goal}$$

policy $\rightarrow$ move right

$$G_0 = (-1) + \gamma(-1) + \gamma^2(-1) + \gamma^3(10)$$

case A:

$$P(S' | s, a) \text{ given}$$

$$R(s, a, s')$$

case B

down to actual
 R_s

you can move
 right or left
 but won't tell
 what happens

It may experience

but won't see
what happens

It may experience

$G_0^{(1)}, G_0^{(2)}, G_0^{(3)}$

$$V^{\pi}(s) \approx E(G)$$
$$= \frac{1}{N} \sum_{i=1}^N G_0^{(i)}$$

Not all possible
future known

So,

80% move right

20% don't go, explore

G_t is RV
random variable

$$V^{\pi}(s) = E_{\pi}[G_t | S_t = s]$$

why E_{π} contains π

$$V^{\pi}(s) = E_{\pi}(G_t | S_t = s)$$

↑
following π policy

Starting from s

Bellman

$$V^{\pi}(s) = E_{\pi}[R_{t+1} +$$
$$\gamma V^{\pi}(s_{t+1}) | S_t = s]$$

S_0, S_1, S_2, S_3, S_4

it always move right

$$V^{\pi}(S_0) = -1 + 0.9 V^{\pi}(S_1)$$

$$V^{\pi}(S_1)$$

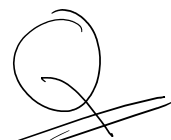
$$V^{\pi}(S_2) = -1 + 0.9 V^{\pi}(S_3)$$

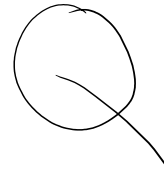
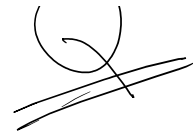
$$V^{\pi}(S_3) = 10 + 0.9 V^{\pi}(S_4)$$

$$V^{\pi}(S_4) = 0$$

because
no future reward
after reaching goal

using this find
all

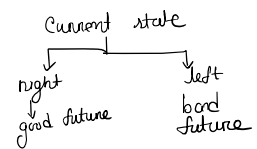




$$V^{\pi}(s) = E_{\pi}[G_t | S_t = s]$$

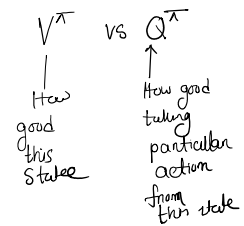
how good is it
to be in state s
if follow policy π

$V^{\pi}(s)$ doesn't tell
which action good.



$$Q^{\pi}(s, a) = E_{\pi}[G_t | S_t = s, A_t = a]$$

expected future
return
if current state = s
and I take action = a
following policy π



eg:

S_0, S_1, S_2, S_3, S_4
goal
action = $\{0, 1\}$
left right



$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$Q^\pi(s, a) = E_\pi[G_t | S_t = s, A_t = a]$$

$$= E_\pi[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s, A_t = a]$$

$$\text{Now, } G_t = R_{t+1} + \gamma G_{t+1}$$

$$Q^\pi(s, a) = E_\pi[G_t | S_t = s, A_t = a]$$

$$= E_\pi[R_{t+1} + \gamma G_{t+1} | s, a]$$

$$E_\pi(G_{t+1} | S_{t+1}) = V^\pi(S_{t+1})$$

Bellman for Q

$$Q^\pi(s, a) = E_\pi[R_{t+1} + \gamma V^\pi(S_{t+1}) | S_t = s, A_t = a]$$

$$\begin{matrix} S_0 & S_1 & S_2 & S_3 & S_4 \\ \gamma = 0.9 \end{matrix}$$

$$V^\pi(S_3) = 10$$

$$Q^\pi(S_2, \text{right}) =$$

$$E_\pi[R_3 + \gamma V^\pi(S_1) | s, a]$$

$$= -1 + 0.9 \times 10$$

$$= 8$$

if

$$Q^\pi(S_2, \text{right}) = 8$$

$$Q^\pi(S_2, \text{left}) = 2$$

Right better than
result

$$Q^* = \arg\max_a Q^\pi(s, a)$$

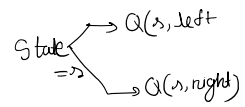
$$Q^* = \arg\max_a (2, 8) = \text{Right}$$

$$V^\pi(s) \text{ relation with } Q$$

policy chosen on
action according to
 $\pi(a|s)$

$$\rightarrow Q(s, \text{left})$$

action $a \in \mathcal{A}$
 $\pi(a|s)$



$$V^\pi(s) = \sum \pi(a|s) Q^\pi(s, a)$$

$$Q^\pi(s, a) = E_\pi(G_t | s, a)$$

$$V^\pi(s) = E_\pi(G_t | s)$$

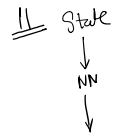
$$P(A|B) = \frac{P(A, B)}{P(B)}$$

$$P(A, B) = P(A|B) P(B)$$

$$E_\pi(G_t | s) = E_\pi(G_t | s, a) \pi(a|s)$$

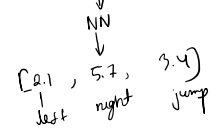
$$V^\pi(s) = \sum \pi(a|s) Q^\pi(s, a)$$

10



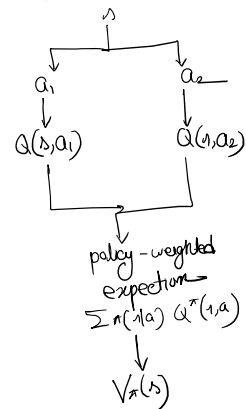
$Q(s, a_1)$ $Q(s, a_2)$ $Q(s, a_3)$

eg
 state = [pos, velocity ...]



agent choose
 action by
 $a = \arg\max_a Q(s, a)$

12



Question

$$p(\text{right} | s_1) = 0.7$$

$$p(\text{left} | s_2) = 0.2$$

$$Q^\pi(s_2, \text{right})$$

$$Q^\pi$$

$$V^\pi(s) = \sum \pi(a|s) Q^\pi(s, a)$$

prove

$$V^\pi(s) = E_n(G_t | S_t = s)$$

$$\pi(a|s) = P(A_t = a, S_t = s)$$

$$\begin{aligned} & \text{So,} \\ & \sum_{A_t=a|S_t=s} E_n(G_t | S_t=s, A_t=a) \\ & = E_n(G_t | s) \end{aligned}$$

$$V^\pi(s) = \sum_a \pi(a|s) E_n(G_t | S_t=s, A_t=a)$$

$$V^\pi(s) = \int \pi(s|a) Q^\pi(s, a) da$$

State vs
Observation

State

Observation $\rightarrow$
agent senses from
environment

State $\rightarrow$ robot
position, velocity
obstacle pos, velocity
battery level

Observation $\rightarrow$ camera
image,
IMU measurements,
lidar measurement

$O_t \neq S_t$
Observation

Martov
property

Markov property

$$H_t = \text{history}$$

$$= (S_0, A_0, R_1, S_1, A_1, R_2, \dots)$$

Markov

$$P(S_{t+1} | S_t, A_t, H_t) \\ = P(S_{t+1} | S_t, A_t)$$

no need of history if S_t and A_t is known

$$S_t = (p, v) \\ = (100, 20) \\ \text{apply } A_t = \text{acceleration}$$

to predict next state, current state is sufficient, we don't need previous state

*) Markov doesn't tell future deterministically.

It is actually random

Transition

Probability

$$P(s' | s, a)$$

if environment stochastic

```

    graph LR
      S2((S2)) -- "0.8" --> S2_2((S2))
      S2 -- "0.2" --> S1((S1))
  
```

$$\sum P(s' | s, a) = 1$$

$$P(s' | s, a)$$

after taking action = a at state = s , what will be the next state

$$P(\text{right} | s_2) = 0.7$$

$$\begin{aligned}
 \pi(\text{Right} | s_2) &= 0.7 \\
 \pi(\text{Left} | s_2) &= 0.3
 \end{aligned}$$

$$\underline{\underline{V^2}} \\
 V^*(s) = \sum_a \pi(a|s) Q^*(s,a)$$

Now

$$\begin{aligned}
 &Q^*(s,a) \\
 &= E [R_{t+1} + \gamma V^*(s_{t+1}) | s, a] \\
 &= \sum_{s'} P(s' | s, a) \{ R(s, a, s') + \gamma V^*(s') \}
 \end{aligned}$$

Now,

$$\begin{aligned}
 V^*(s) &= \sum_a \pi(a|s) Q^*(s,a) \\
 &= \sum_a \pi(a|s) \times \sum_{s'} P(s' | s, a) \{ R(s, a, s') + \gamma V^*(s') \}
 \end{aligned}$$

Ques

Stochastic environment

a robot
command = right
20% chance it explodes

